

# 20250318EB\_MN (MS, qPCR)

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2025年3月18日 星期二

Use **Neurobasal-A without glucose and sodium pyruvate** for experiment. I added

**0.5mM Na Pyruvate** this time.

One well of cell still died. However, from Plate B and C, cells can survive with either

**0.5mM Na pyruvate** or **N2+B27**

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年3月20日 星期四

T2 -> T3

2025年3月22日 星期六

T3 -> T4

2025年3月24日 星期一

T4 -> T5

2025年3月26日 星期三

T5 -> T6

2025年3月28日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年3月31日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatant (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

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2025年4月2日 星期三

T7-> T7

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2025年4月3日 星期四

T7 -> BP T8

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2025年4月4日 星期五

700uL BP T8 -> 800 uL BP T8

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2025年4月7日 星期一

700uL BP T8 -> 800 uL BP T8

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2025年4月9日 星期三

700uL BP T8 -> 800 uL BP T8

treat plate B&C B3 with 0.15mM glucose in Neurobasal-A w/o glu and pyruvate

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2025年4月10日 星期四

## LC-MS treatment at 11:10 am

- I realized that B3 of plateB&C died, so I added additional Na Pyruvate (to around 0,5 mM) to each wells. For A1 and B1, they have 2mL media, I took out 10uL media and added 10uL 100mM pyruvate to each well. For A2,A3,B2,B3, they have around 2.15mL/well. Therefore, I took out 10.7uL media from each well and added 10.7uL 100mM Na Pyruvate into each well. I pipetted the media gently inside the well after adding Na pyruvate.

plate A (with 0.5mM pyruvate)

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|   | 1              | 2              | 3 |
|---|----------------|----------------|---|
| A | 2.5 mM C12-glu | 2.5mM C13-glu  |   |
| B | 0.15mM C12-glu | 0.15mM C13-glu |   |

QPCR 24h treatment at 7:00 pm

- Cells died in B3 within 24h --> same as 250310EB\_MN. Therefore, I decided to test if cells survive with 0.5mM Na Pyruvate and/or with N2/B27 supplements. We never added N2 and B27 in hypoglycemia medium, but BrainPhys has N2 B27 supplements in it.

plateB

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|   | 1 | 2 | 3    |
|---|---|---|------|
| A | 1 | 2 | 9    |
| B | 3 | 4 | died |

plateC

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|   | 1 | 2 | 3    |
|---|---|---|------|
| A | 5 | 6 | 10   |
| B | 7 | 8 | died |

media preparation

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|   | A                     | B           | C                  | D         | E                            |
|---|-----------------------|-------------|--------------------|-----------|------------------------------|
| 1 |                       | volumn (mL) | B27 w/o vitA (50x) | N2 (100x) | Na Pyruvate (100mM to 0.5mM) |
| 2 | Neurobasal-A + 2.5mM  | 6           | 120                | 60        | 30                           |
| 3 | Neurobasal-A + 0.15mM | 6           | 120                | 60        | 30                           |

| Table1 |    |                       |         |                   |                |           |
|--------|----|-----------------------|---------|-------------------|----------------|-----------|
|        | A  | B                     | C       | D                 | E              | F         |
| 1      |    | glucose concentration | N2, B27 | 0.5mM Na Pyruvate | treatment time |           |
| 2      | 1  | 2.5                   | v       | v                 | 1h             |           |
| 3      | 2  | 0.15                  | v       | v                 | 1h             |           |
| 4      | 3  | 2.5                   | v       | v                 | 24h            |           |
| 5      | 4  | 0.15                  | v       | v                 | 24h            |           |
| 6      | 5  | 2.5                   |         | v                 | 1h             |           |
| 7      | 6  | 0.15                  |         | v                 | 1h             |           |
| 8      | 7  | 2.5                   |         | v                 | 24h            |           |
| 9      | 8  | 0.15                  |         | v                 | 24h            |           |
| 10     | 9  | 2.5                   | v       |                   | 24h            |           |
| 11     | 10 | 0.15                  | v       |                   | 24h            |           |
| 12     | 11 | 0.15                  |         |                   | 24h            | cell died |

2025年4月11日 星期五

LC-MS extraction

- plate A B1 died. (I don't know why since I added Na pyruvate to all of them)

| LC-MS condition |                        |                       |         |             |                |      |       |                 | / |
|-----------------|------------------------|-----------------------|---------|-------------|----------------|------|-------|-----------------|---|
|                 | A                      | B                     | C       | D           | E              | F    | G     | H               |   |
| 1               |                        | glucose concentration | glucose | Na Pyruvate | treatment time | cell | media | note            |   |
| 2               | A                      | 2.5                   | C12     | v           | 24h            | v    | v     |                 |   |
| 3               | B                      | 2.5                   | C13     | v           | 24h            | v    | v     |                 |   |
| 4               | C                      | 2.5                   | C13     | v           | 24h            | v    | v     |                 |   |
| 5               | D                      | 0.15                  | C12     | v           | 24h            |      | v     | cell died       |   |
| 6               | E                      | 0.15                  | C13     | v           | 24h            | v    | v     |                 |   |
| 7               | F                      | 0.15                  | C13     | v           | 24h            | v    | v     |                 |   |
| 8               | 0mM media              | 0                     | C12     |             | 24h            |      | v     |                 |   |
| 9               | 0.15mM media +pyruvate | 0.15                  | C12     | v           | 24h            |      | v     | have triplicate |   |
| 10              | 2.5mM media +pyruvate  | 2.5                   | C12     | v           | 24h            |      | v     | have triplicate |   |

QPCR (cont')

- all the condition from 1-10 survived. --> cells can survive with N2B27 or 0.5mM Na pyruvate alone (although not sure what happen with plate A B1. Is the concentration a bit off?)
- treat 1,2,5,6 at 6pm
- collect cells with 1mL TRIzol at 7pm
  - Wash cells with cold PBS
  - Add 1mL trizol to each well
  - incubate at RT for ~5min
  - collect cells into 1.5mL tube
  - store sample in -80C